

Foundations of Data Science

Lecture 2: Data and exploratory analysis

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Data



What is data?

Data are **recorded observations** about the world.

They can take many forms, including numbers, text, images, and more.



A running example

We will follow one dataset through this lecture: the **Palmer Archipelago penguins**, 344 birds of three species (Adelie, Chinstrap, Gentoo), measured on three islands between 2007 and 2009.

Someone stood in the cold, caught a penguin, measured its bill with a caliper and put it on a scale. What follows is about what such records are, and what they are not.

Mathematically, data can be viewed as a function f that maps real-world entities ω to measurable values x ,

$$f : \Omega \rightarrow \mathcal{X},$$

where

- Ω is the sample space of possible states ω of the world,
- $\mathcal{X}$ is the measurement space of possible observations x .

Examples:

- Penguin body mass: $\omega \in \{\text{penguins}\} \rightarrow x \in \mathbb{R}^+$ (g)
- Stock price: $\omega \in \{\text{market states}\} \rightarrow x \in \mathbb{R}^+$ (USD)
- Pixel colour: $\omega \in \{\text{scenes}\} \rightarrow x \in \{0, \dots, 255\}^3$ (RGB values)

If the sample space Ω carries a probability distribution $p(\omega)$, then f turns a random state of the world into a random observation $x = f(\omega)$. Its distribution is the one f induces from p , called the **data distribution** $p_r(x)$, where r stands for "real".

When densities exist, it can be written as

$$p_r(x) = \int_{\omega \in \Omega} p(\omega) \delta(x - f(\omega)) d\omega,$$

where δ is the Dirac delta function.

The **measurement process** is part of the data generation mechanism. We make it explicit by adding the measurement conditions θ (instrument settings, environmental conditions, observer effects) to the map,

$$f : \Omega \times \Theta \rightarrow \mathcal{X}.$$

Measurements can introduce quantization (continuous to discrete), noise (random perturbations), and bias (systematic deviations). If $\Omega \times \Theta$ carries a joint distribution $p(\omega, \theta)$, then

$$p_r(x) = \iint_{\omega \in \Omega, \theta \in \Theta} p(\omega, \theta) \delta(x - f(\omega, \theta)) d\omega d\theta,$$

which captures the variability of both the phenomenon and its measurement.



Example: how a penguin record is made.

- Bill length and depth: dial calipers, to 0.1 mm.
- Flipper length: ruler, to 1 mm.
- Body mass: Pesola spring scale and a weigh bag, to 25 g.
- Sex: not measured at all, but inferred in the lab from a blood sample.

Nests are found at the one-egg stage, both adults are caught, measured, sampled and released. All of this is θ .

The instruments are visible in the data: every body mass is a multiple of 25 g, every bill length has one decimal, every flipper length is a whole millimetre.

Data types

Atomic data are the indivisible units of information collected through measurements. They are often categorized based on their nature and the operations that can be performed on it.

- Numerical (continuous, discrete)
- Categorical (nominal, ordinal)

Numerical data

- Continuous: $x \in \mathbb{R}$ (e.g., temperature), $x \in \mathbb{R}^+$ (e.g., height, weight)
- Discrete: $x \in \mathbb{N}$ (e.g., counts), $x \in \mathbb{Z}$ (e.g., differences)

Categorical data

- Nominal: $x \in \mathcal{C} = \{c_1, c_2, \dots, c_n\}$ (e.g., colors, types, text characters) without intrinsic order
- Ordinal: $x \in \mathcal{C}$ with ordering relations $c_1 \prec c_2 \prec \dots \prec c_n$ (e.g., ratings, grades)



Example: one penguin record

species: Adelie	# Categorical, nominal
island: Torgersen	# Categorical, nominal
bill length: 39.1 mm	# Numerical, continuous
body mass: 3750 g	# Numerical, continuous
sex: male	# Categorical, nominal
year: 2007	# Numerical, discrete

No variable here is ordinal; a rating from 1 to 5, or a grade, would be.

Data structures

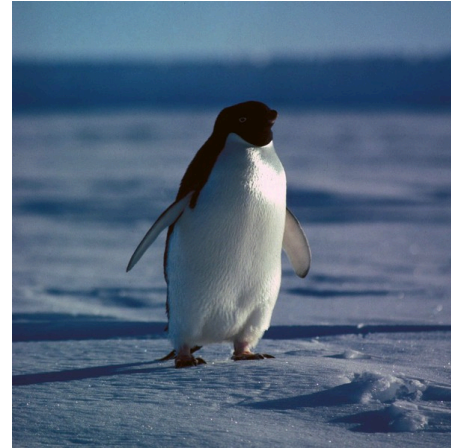
A measurement x can be a single atomic value or a composite structure made of multiple atomic values. Common aggregates or data structures include:

- Tabular data
- Arrays and tensors
- Sequences
- Networks and graphs

Tabular data

	species	island	bill length (mm)	body mass (g)	sex
0	Adelie	Torgersen	39.1	3750.0	male
1	Adelie	Torgersen	39.5	3800.0	female
2	Adelie	Torgersen	40.3	3250.0	female
3	Adelie	Torgersen	NaN	NaN	NaN
4	Adelie	Torgersen	36.7	3450.0	female
...	...	...	...	...	...
339	Chinstrap	Dream	55.8	4000.0	male
340	Chinstrap	Dream	43.5	3400.0	female
341	Chinstrap	Dream	49.6	3775.0	male
342	Chinstrap	Dream	50.8	4100.0	male
343	Chinstrap	Dream	50.2	3775.0	female

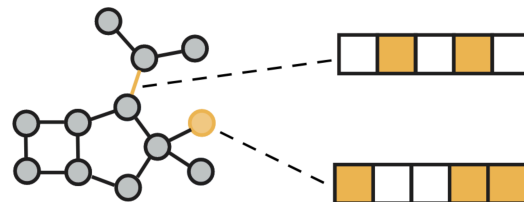
Arrays and tensors



Sequences

```
['F', 'l', 'o', 'w', 'e', 'r']
```

Networks and graphs



A data frame $\mathbf{X}$ represents a **tabular collection** of n records (rows) over d variables/atomic measurements (columns),

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1d} \\ x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nd} \end{pmatrix}.$$

Each entry x_{ij} corresponds to the value of variable j for record i .

Variables are often heterogeneous (mixing numerical and categorical types). When all variables are numerical, the data frame can be viewed as a matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$.

For the penguins, $n = 344$ records over $d = 8$ variables, four of them numerical.

Collections of homogeneous measurements can be represented as **arrays** or **tensors** $\mathbf{X} \in \mathbb{R}^{d_1 \times d_2 \times \cdots \times d_k}$, where the position of each atomic value in the array is usually associated to a spatial or temporal location.

- Images: 3d arrays $\mathbf{X} \in \{0, \dots, 255\}^{h \times w \times c}$ (height, width, channels). The photo on the previous slide is such an array.
- Videos: 4d arrays $\mathbf{X} \in [0, 255]^{t \times h \times w \times c}$ (time, height, width, channels).

Data can also be structured as ordered **sequences** $S = (x_1, x_2, \dots, x_T)$ indexed by time or position. Each element x_t can be atomic or composite.

- Time series: $S = (x_1, x_2, \dots, x_T)$ where x_t is a measurement at time t (e.g., stock prices).
- Text: $S = (w_1, w_2, \dots, w_T)$ where w_t is the t -th word in a document.

Finally, data can be organized as **networks** or **graphs** $G = (V, E)$, where entities are represented as nodes V and relationships as edges E . Each may also have associated attributes, x_v for nodes and x_{uv} for edges.

- Molecular structures, where nodes represent atoms and edges represent bonds.
- Social networks, where nodes represent individuals and edges represent interactions or relationships.



Follow the tutorials in `nb02a-tables.ipynb` , `nb02b-jax.ipynb` , and `nb02c-data-wrangling.ipynb` to practice working with arrays and data frames in Python.

Data quality

Real-world data are often imperfect and may suffer from various **quality issues** that can impact analysis and modeling. Common data quality issues include:

- Missing values
- Measurement errors
- Outliers

Missing values are common in real-world datasets and can arise from various factors such as non-response in surveys, sensor malfunctions, or data corruption.

Let $\mathbf{X}_{\text{full}}$ be the complete data and $\mathbf{M} \in \{0, 1\}^{n \times d}$ the missingness pattern, with $m_{ij} = 1$ when entry ij is observed and $m_{ij} = 0$ when it is missing¹.

What we actually hold is the pair $(\mathbf{X}_{\text{obs}}, \mathbf{M})$, where $\mathbf{X}_{\text{obs}} = \{x_{ij} : m_{ij} = 1\}$ are the observed entries and $\mathbf{X}_{\text{mis}} = \{x_{ij} : m_{ij} = 0\}$ the missing ones. A missing entry is not a zero, and no arithmetic can recover it.

Two penguins were never measured, and eleven have no recorded sex.

1: Rubin's convention is the opposite, 1 for missing.

The pattern $\mathbf{M}$ is itself produced by the measurement process, and is modeled as such:

- Missing completely at random (MCAR): the pattern is independent of the data, $p(\mathbf{M} \mid \mathbf{X}_{\text{full}}) = p(\mathbf{M})$.
- Missing at random (MAR): the pattern may depend on what is observed, but not on what is missing, $p(\mathbf{M} \mid \mathbf{X}_{\text{full}}) = p(\mathbf{M} \mid \mathbf{X}_{\text{obs}})$.
- Missing not at random (MNAR): the pattern depends on the missing entries themselves.

Why the distinction matters: what we can compute is

$$p(\mathbf{X}_{\text{obs}}, \mathbf{M}) = \int p(\mathbf{X}_{\text{full}})p(\mathbf{M} \mid \mathbf{X}_{\text{full}})d\mathbf{X}_{\text{mis}}.$$

Under MCAR or MAR, the second factor does not depend on the missing entries and, provided it shares no parameters with the first, it can be ignored: modelling the observed data is enough. Under MNAR it cannot be ignored, and the mechanism must be modeled jointly with the data.

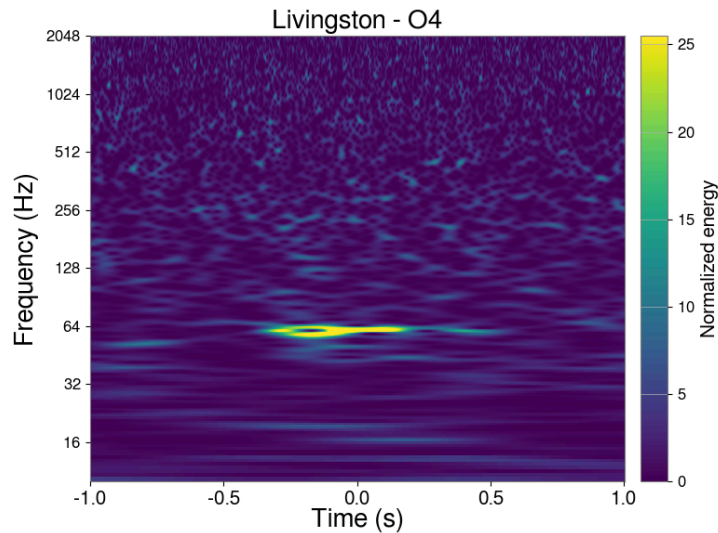
Example:

Survey of $n = 1000$ respondents, 30% do not answer the income question.

- MCAR: Randomly selected respondents skip the question.
- MAR: Younger respondents are less likely to answer.
- MNAR: High earners refuse to answer.

Imputing or discarding missing values without making explicit the assumptions about the missingness mechanism can lead to biased results.

The measurement process can also introduce acquisition errors and produce observations that deviate significantly from regular measurements. These **outliers** can arise from instrument malfunctions, data entry errors, or rare events.



Example:

Glitches in gravitational wave detectors are outliers that can mimic true signals and complicate detection efforts. They can arise from environmental disturbances, instrumental artifacts, or other non-astrophysical sources.

Treating outliers requires a model of the measurement process that either describes measurements under normal conditions or explicitly accounts for anomalies.

Two models make this explicit: a contamination mixture

$$p(x) = (1 - \varepsilon)p_{\text{model}}(x) + \varepsilon p_{\text{bad}}(x),$$

which gives bad measurements their own distribution, or a heavy-tailed data model, which allows rare large deviations without special-casing them. We fit the second to Newcomb's speed-of-light measurements in Lecture 7.

Outliers should not be removed blindly unless explicitly justified by the measurement model or domain knowledge.



Temperature readings in Liège:

20.1, 19.8, 20.3, 1000.0, 20.2, 19.9

The 1000.0 value is an outlier likely due to a sensor error.

19.2, 19.8, 20.3, 37.8, 20.2, 19.9

The 37.8 value is a rare but plausible measurement on a hot day.

Where this leaves us

Everything in this part is one picture of how the data came to be:

- the map f from states of the world to measurements,
- the conditions θ under which measurements are taken,
- the pattern $\mathbf{M}$ of what ends up recorded,
- the data distribution $p_r(x)$ that all of this induces.

In `nb01`, `simulate()` was such a map f , its Gaussian noise played the role of θ , and the landing distance was x . The generative model ran forward, and inference ran it backward.

From Lecture 4 onwards, we build parametric models of $p_r(x)$ and fit them to data. Before that, we look at what the data themselves show.

Exploratory data analysis

Exploratory data analysis (EDA) is the systematic examination of data to understand its structure, patterns and anomalies before formal modeling. It typically involves visual and quantitative techniques to summarize key characteristics of the data.

The goal is to generate hypotheses and inform modeling decisions, not to confirm preconceived notions.



Back to the penguins, with the whole table in hand: 344 records, four numerical measurements, three species and three islands.

Switch to `nb02d-eda.ipynb` to follow along.

Univariate analysis

Let us consider a variable j from a data frame $\mathbf{X} \in \mathbb{R}^{n \times d}$, represented as the vector $\mathbf{x}_j = (x_{1j}, x_{2j}, \dots, x_{nj})^T$.

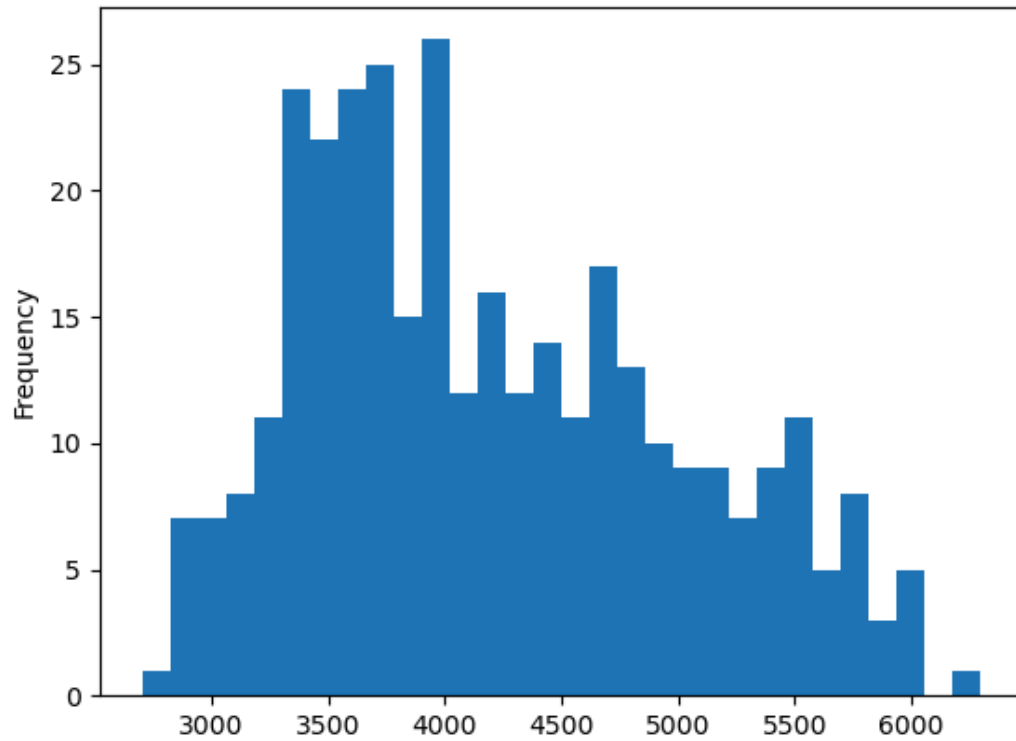
Univariate analysis focuses on understanding the distribution and characteristics of this single variable.

For numerical variables, common techniques include:

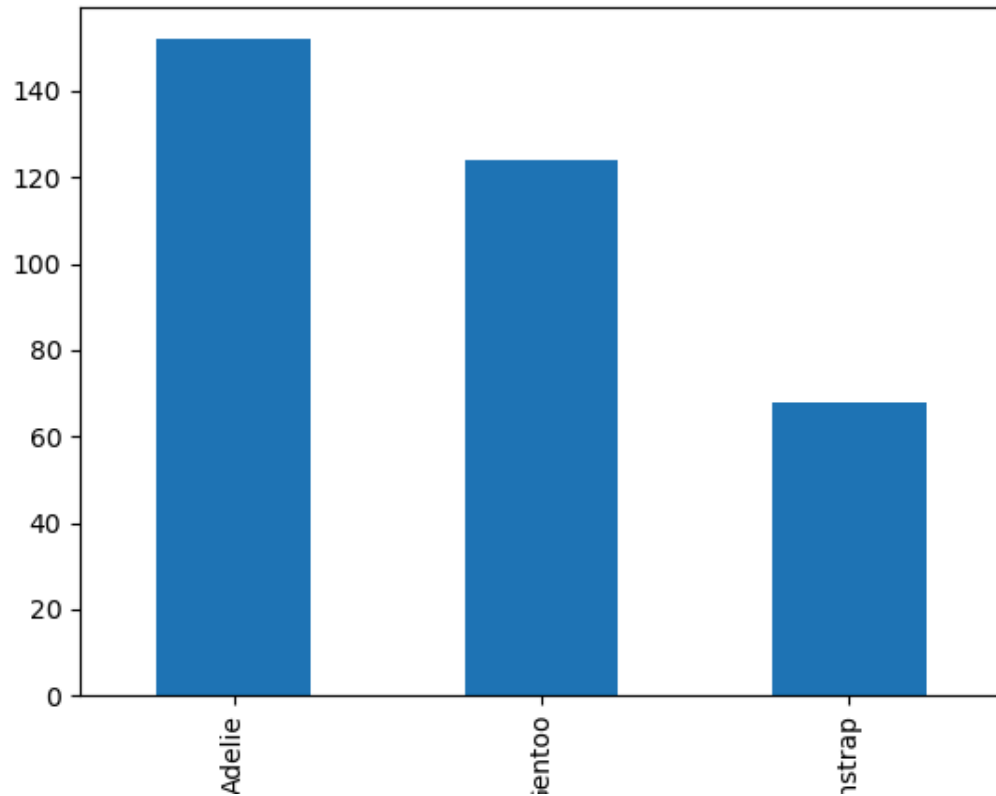
- Looking at the raw data: print values, scroll through them.
- Plotting the data: histograms reveal the distribution shape.
- Summarizing with statistics: mean, median, mode, variance, skewness, kurtosis.

For categorical variables:

- Counting occurrences of each category.



Numerical: Histograms of body mass for all penguins.



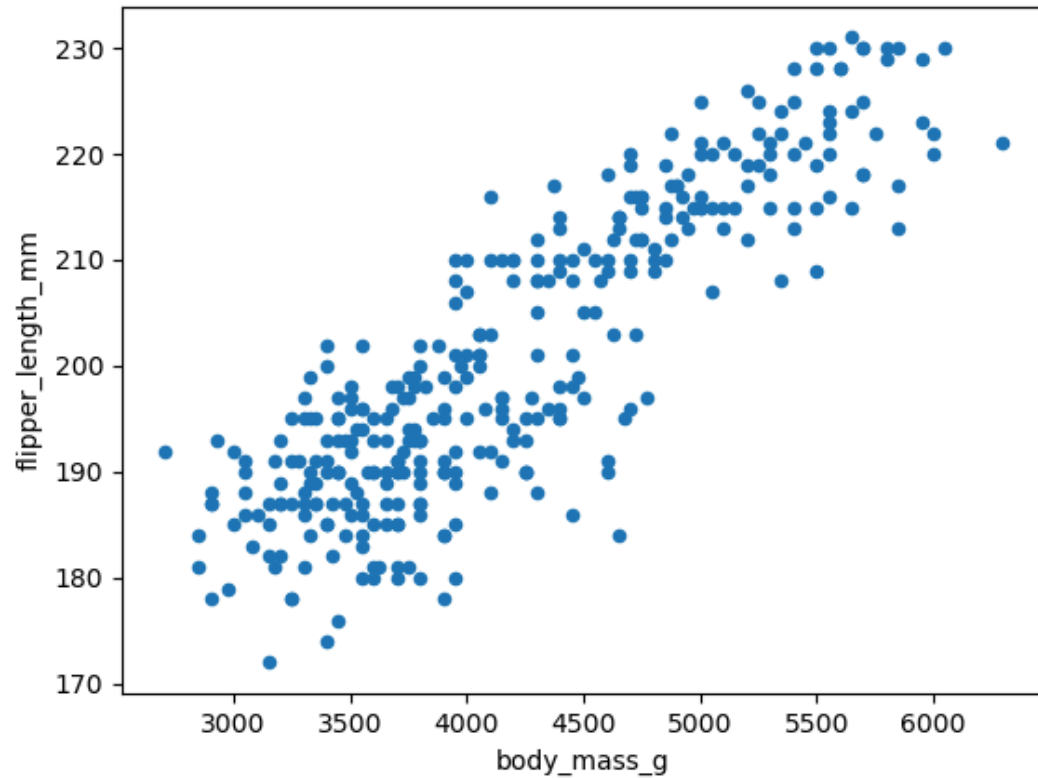
Categorical: Bar plot of species counts for all penguins.

Bivariate analysis

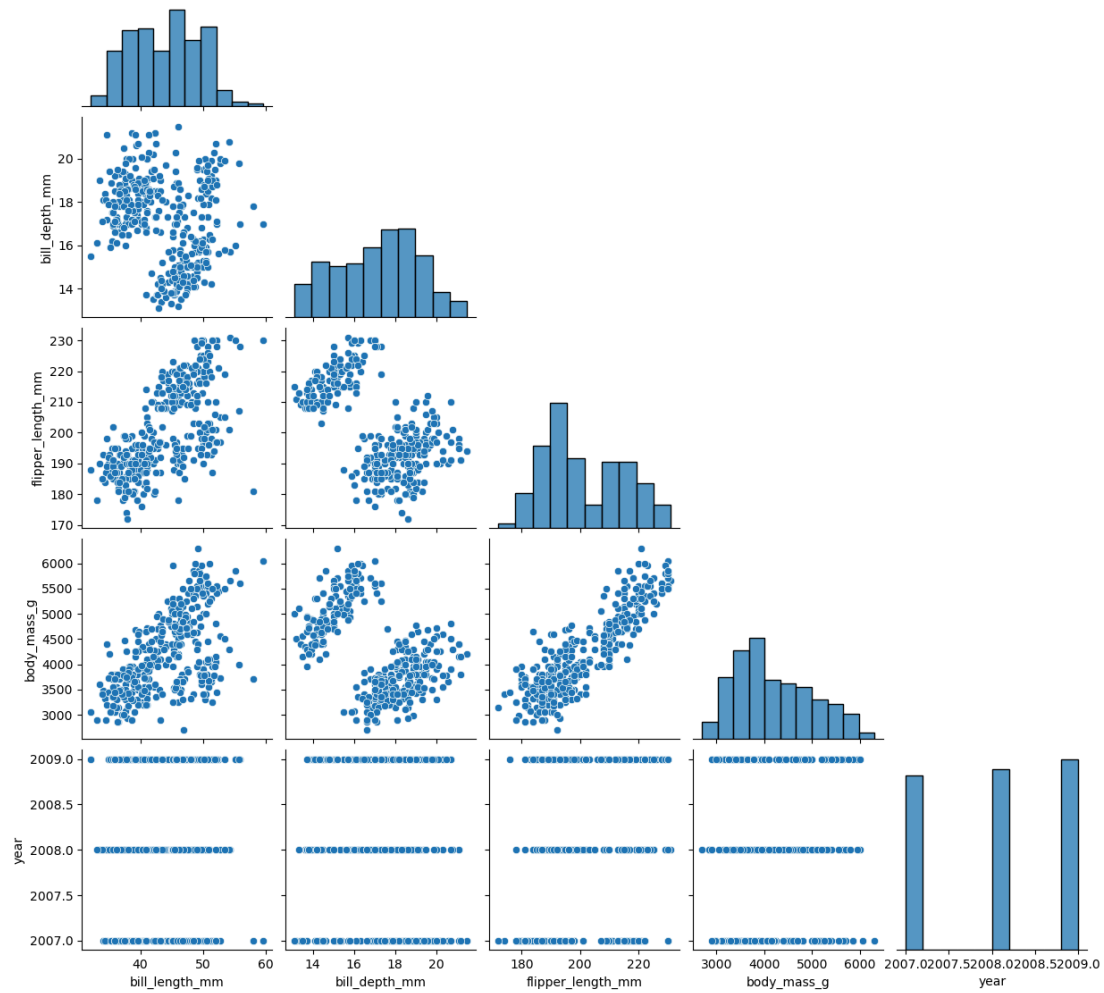
Bivariate analysis examines the relationship between two variables j and k from a data frame $\mathbf{X} \in \mathbb{R}^{n \times d}$, represented as the vectors $\mathbf{x}_j$ and $\mathbf{x}_k$.

Depending on the types of variables, different techniques are used:

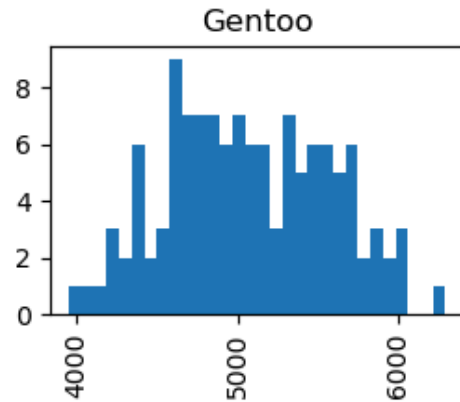
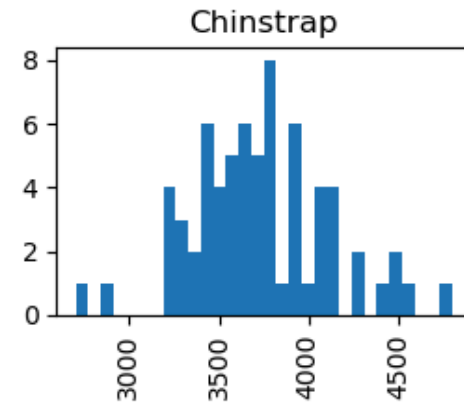
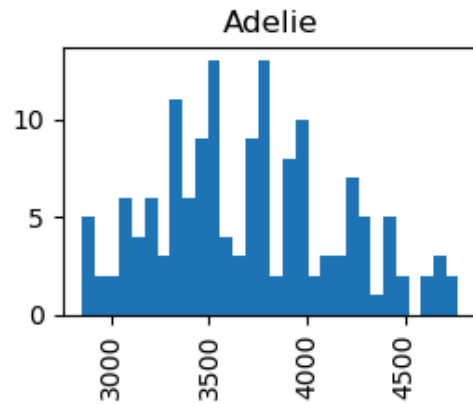
- Pair plots for two numerical variables (scatter or 2d histogram).
- Multiple histograms for categorical vs numerical variables.
- Contingency tables for two categorical variables.
- Correlation coefficients (e.g., Pearson, Spearman) for numerical variables.



Numerical vs. numerical: Scatter plot of body mass vs flipper length.



Pair plots of all numerical variables.



Categorical vs. numerical: Histograms of body mass by species.

island	Biscoe	Dream	Torgersen
species			
Adelie	44	56	52
Chinstrap	0	68	0
Gentoo	124	0	0

Categorical vs. categorical: Contingency table of species and island.

Correlation coefficients can quantify the dependency between two numerical variables. They are useful but come with assumptions and limitations.

- Pearson correlation measures linear relationships. Its empirical version reads

$$\hat{\rho}_{jk} = \frac{\sum_{i=1}^n (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k)}{\sqrt{\sum_{i=1}^n (x_{ij} - \bar{x}_j)^2} \sqrt{\sum_{i=1}^n (x_{ik} - \bar{x}_k)^2}},$$

an estimate of $\rho_{jk} = \text{cov}(x_j, x_k) / (\sigma_j \sigma_k)$. It ignores non-linear dependencies.

- Spearman correlation is the Pearson correlation of the rank-transformed variables. It captures monotonic relationships.
- Correlation does not imply causation and can be affected by outliers.

The **mutual information** between two variables x_j and x_k measures the reduction in uncertainty about one given knowledge of the other. For discrete variables,

$$I(x_j; x_k) = \sum_{x_j} \sum_{x_k} p(x_j, x_k) \log \frac{p(x_j, x_k)}{p(x_j)p(x_k)},$$

with integrals in place of the sums for continuous ones.

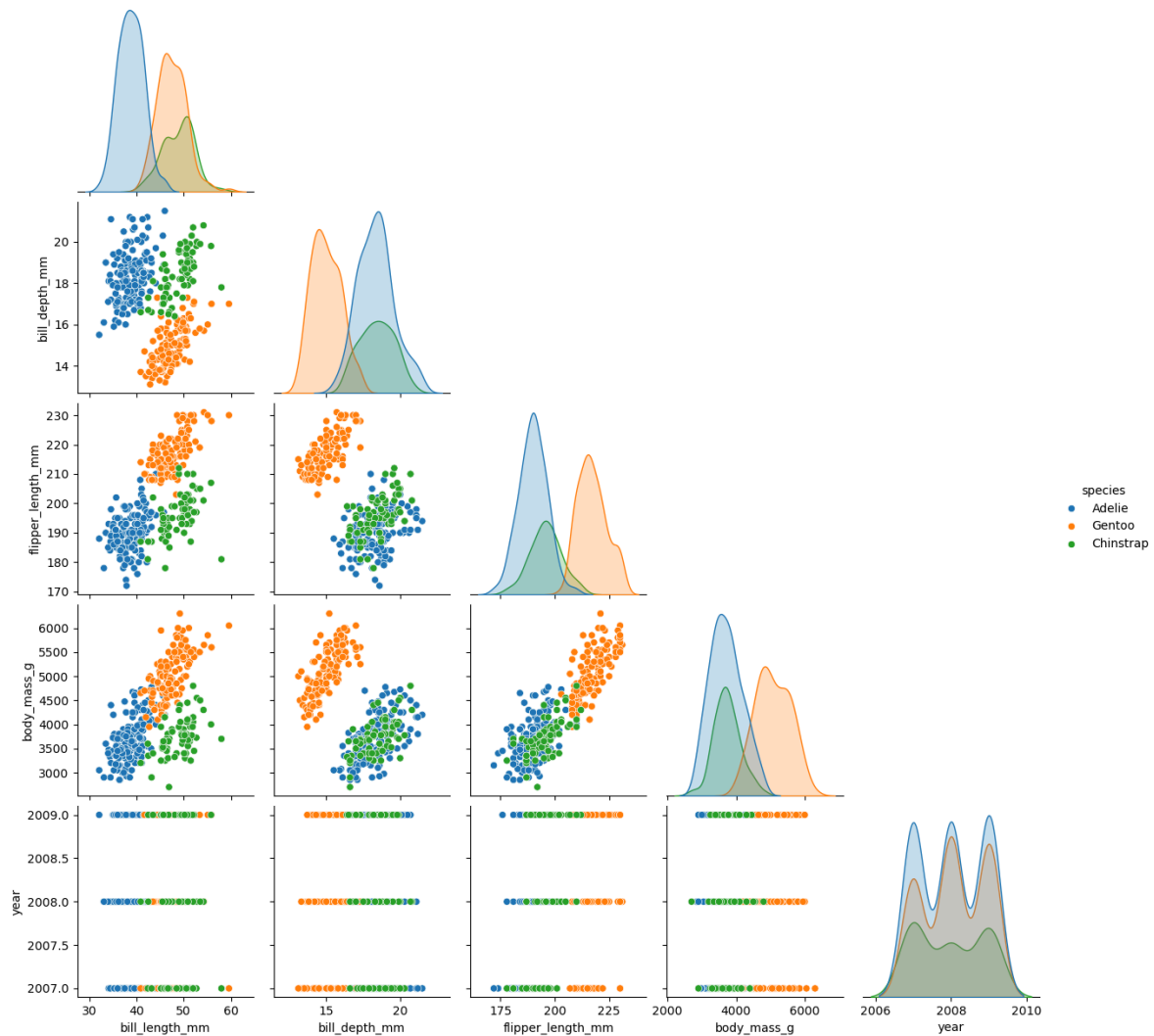
Mutual information captures any statistical relationship, not just linear or monotonic ones. It is defined on the distributions, however, which must themselves be estimated from the n records, and that is hard.

Multivariate analysis

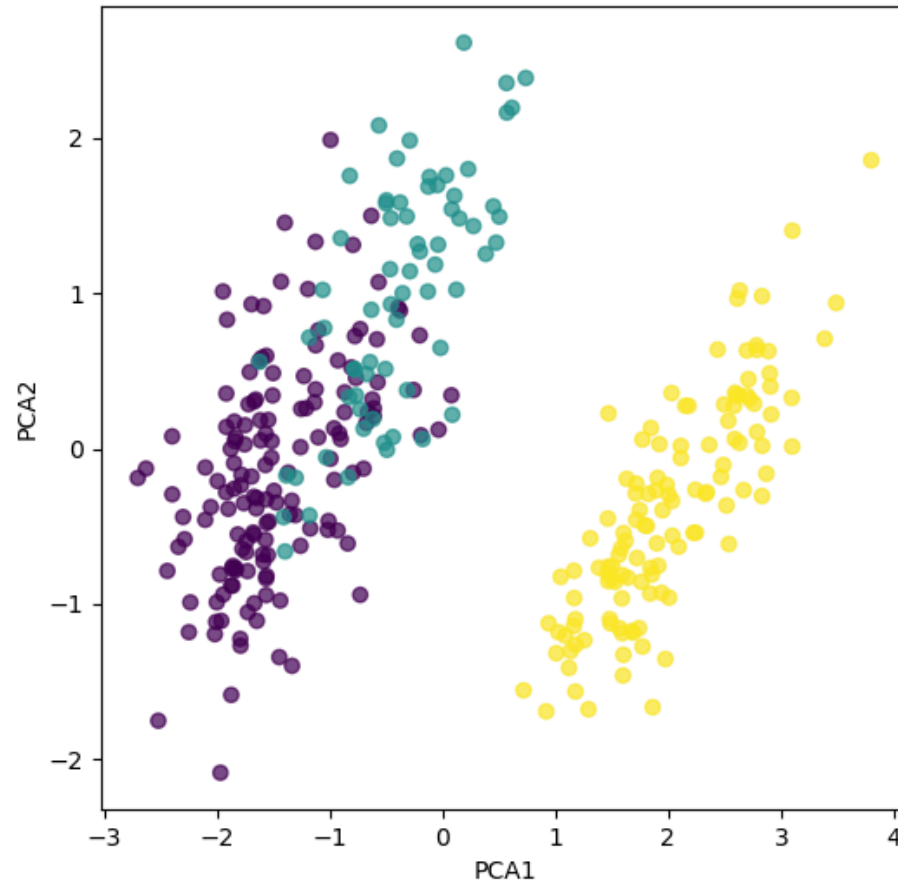
Multivariate analysis explores relationships among three or more variables in a data frame $\mathbf{X} \in \mathbb{R}^{n \times d}$.

Common techniques include:

- The same as bivariate analysis, but conditioning on a third variable (e.g., pair plots colored by species).
- Dimensionality reduction methods (e.g., PCA, t-SNE) to visualize high-dimensional data.
- Clustering algorithms (e.g., k-means, hierarchical clustering) to identify groups of similar records.



Pair plots of all numerical variables, colored by species.



PCA projection of all numerical variables, colored by species.

EDA within Box's loop

EDA is a crucial step in Box's loop for data analysis. It helps to understand the data, generate hypotheses, and **inform modeling decisions**.

- Distribution shapes can suggest appropriate model families.
- Relationships between variables can guide model structure.
- Insights about scale and variance can inform data transformations.

After building a model and computing initial results, EDA can be used to critique the model fit and identify areas for improvement.

- Residual analysis can reveal patterns not captured by the model.
- Prediction errors may point to specific data subsets that are problematic.
- Unexpected patterns in the data can suggest new features or model revisions.

The end.